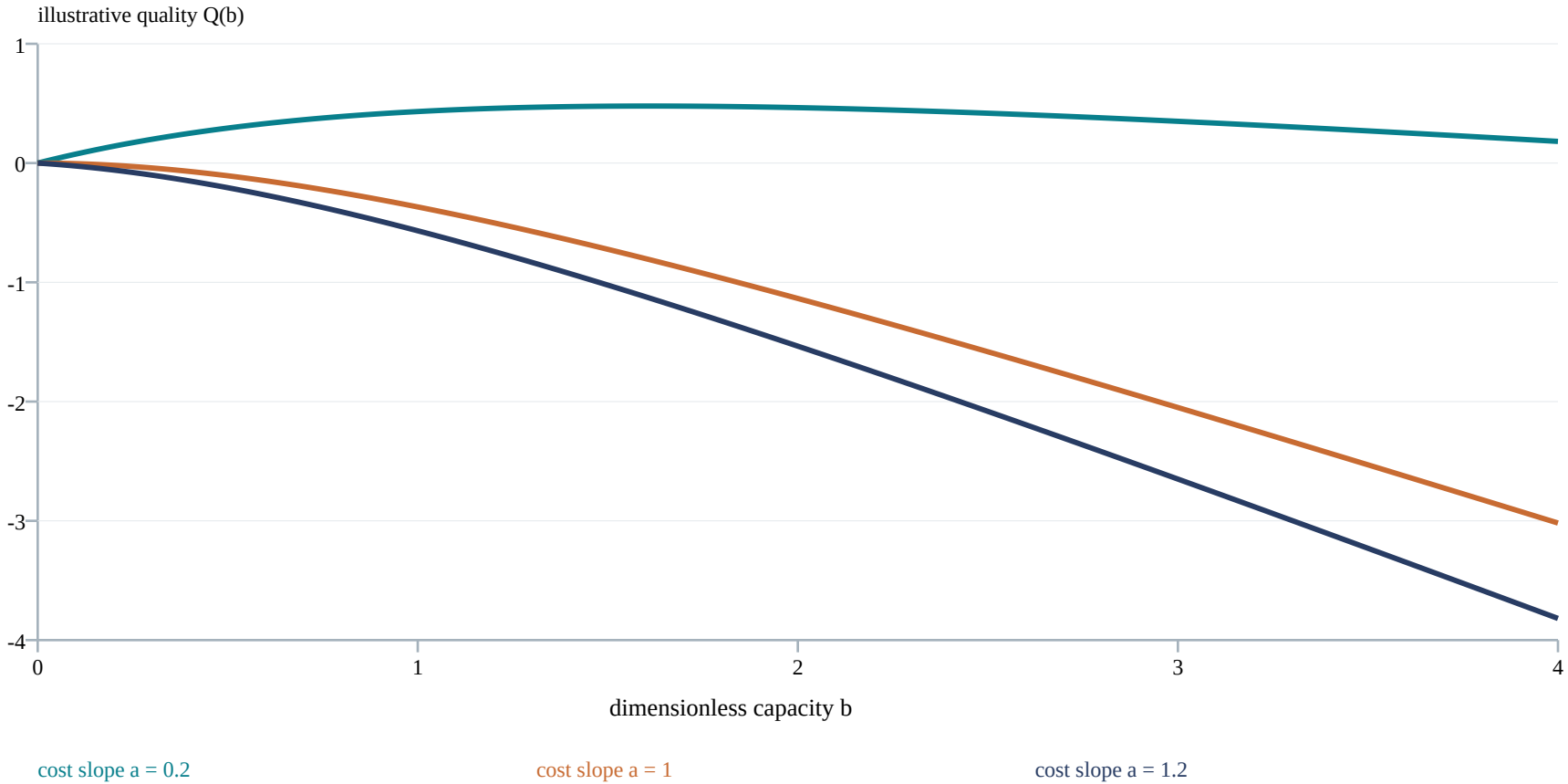
Increasing cost need not yield an interior optimum

Analytic counterexamples: Q(b) = 1 - exp(-b) - a*b; b is dimensionless